Uncertainty Meets Opportunity
Joint Probabilistic Forecasting for Battery Arbitrage in CEE Day-Ahead Power Markets

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Motivation

- European power systems are moving from controllable fossil generation toward more solar and wind.
- This creates a simple daily tension: electricity can be abundant and cheap around midday, then scarce and expensive in the evening.
- Recent shocks in gas, CO_2 and electricity markets have made price spikes and very low-price hours more visible.
- Batteries are attractive because they can buy electricity when the system has surplus and sell it when the system needs flexibility.
- The difficult question is economic: are day-ahead price spreads large and predictable enough to pay for losses, tariffs and the battery investment?

Example. If solar production pushes the price close to zero at 12:00 and evening demand raises the price at 18:00, a battery can charge at 12:00 and discharge at 18:00. The trade is profitable only if the spread remains after efficiency losses and transaction costs.

The four CEE markets

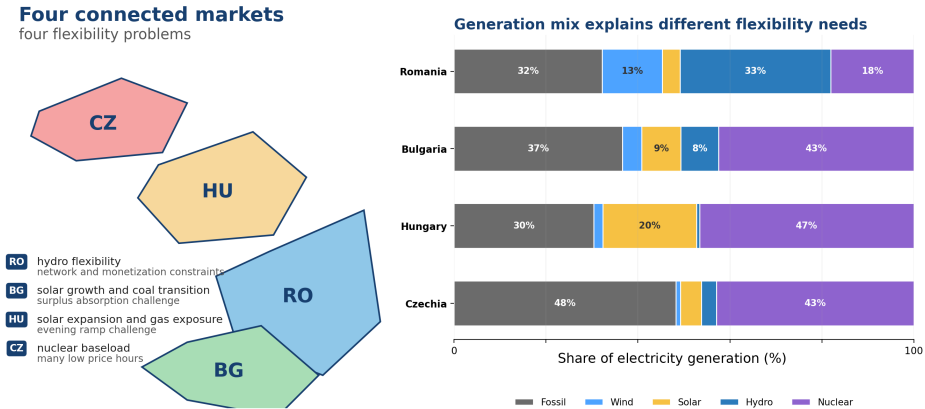
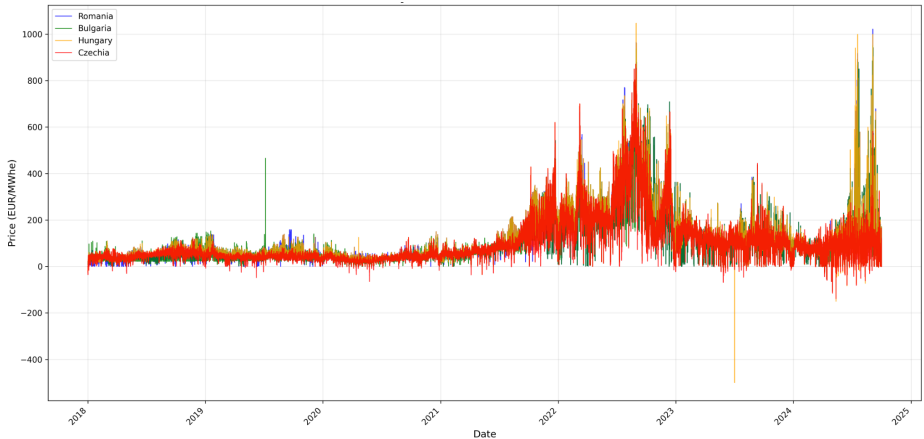


Figure 1. Connected bidding zones and heterogeneous generation mixes.

Romania	hydro flexibility and network limits	Bulgaria	solar growth and coal transition
Hungary	solar expansion and gas exposure	Czechia	nuclear baseload and many low-price hours

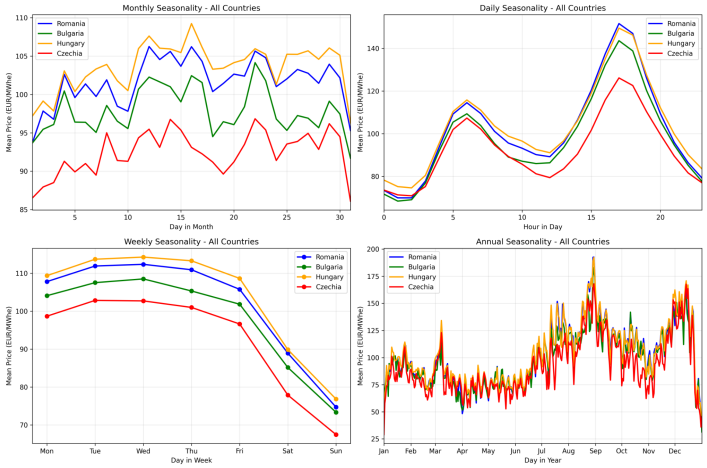
Day-ahead prices across the four markets

Figure 2. Hourly day-ahead prices from January 2018 to September 2024

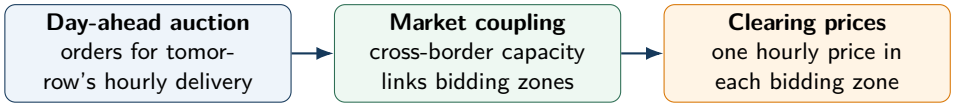


Intraday and seasonal structure

Figure 3. Monthly, hourly, weekly and annual average price patterns



Day-ahead market setting in the paper



Where price differences come from

- If interconnection capacity is sufficient, zonal prices tend to converge.
- If capacity or other frictions bind, prices can separate across countries.

How this enters the paper

- Single-market case: buy and sell inside one bidding zone.
- Multi-market case: choose both the country and the hour.

Important caveat. The multi-market case is a benchmark for regional flexibility value. It is not a claim that all cross-border spreads are fully attainable after tariffs, congestion and transmission-access costs.

What storage-arbitrage studies already show

Study	Market and sample	Storage	Forecast or operation setting	Main result for our narrative
Walawalkar et al. (2007)	NYISO, 2001-2005	NaS, flywheel	Historical market scenarios	Location, efficiency and market product affect revenue capture.
Bradbury et al. (2014)	Seven US markets, 2008	14 technologies	Perfect-foresight optimization	Costs and efficiency determine whether arbitrage reaches target IRR.
Nunez et al. (2022)	24 European countries, 2019	Li-ion	MILP, NPV and IRR	Price dispersion and country risk create heterogeneous feasibility.
Komorowska and Olczak (2024)	22 European day-ahead markets, 2016-2022	Li-ion	Fixed-hour arbitrage and NPV	Current-cost NPV remains negative; lower CAPEX improves selected markets.
Veenstra and Mulder (2025)	Netherlands, 2006-2023	Batteries	Day-ahead and intraday strategies with price effects	Returns can decline as battery capacity grows and changes prices.
O'Connor et al. (2025)	Day-ahead and balancing markets	Simulated BESS	Probabilistic intervals and trading algorithm	Forecast uncertainty should be evaluated through trading outcomes.

Table 1. Representative studies linking storage arbitrage, forecasting and economic evaluation.

Abbreviations: NaS = sodium-sulfur; BESS = Battery Energy Storage System; MILP = mixed-integer linear programming; NPV = Net Present Value; IRR = Internal Rate of Return.

General conclusion. Batteries are useful for flexibility and can earn arbitrage revenue, but the literature does not imply automatic profitability. Feasibility depends on persistent spreads, efficiency, CAPEX, tariffs, market design and complementary revenue streams.

Contribution of this article

1. Forecasting and dispatch under uncertainty

- Estimate full conditional day-ahead price distributions with DRF.
- Translate distributions into quantile-based limit orders.
- Decide orders before realized prices are known.
- Evaluate against fixed-hour and perfect-foresight benchmarks.

2. Unified CEE market comparison

- Compare Romania, Bulgaria, Hungary and Czechia in one framework.
- Study both stand-alone and cross-country arbitrage.
- Identify regional flexibility opportunities from cross-border price differences.
- Treat the multi-market case as a benchmark under cross-market access.

Economic contribution. The article moves from theoretical ex-post arbitrage potential to forecast-based value, making it possible to evaluate realized returns and investor-oriented metrics such as NPV, IRR, payback and break-even CAPEX.

Data and predictor set

Panel: Romania, Bulgaria, Hungary and Czechia **Period:** January 2018 to September 2024 **Target:** hourly day-ahead price.

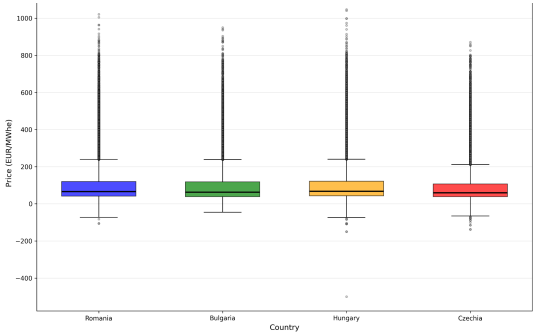
Predictor family in the paper	Count	What it captures
Demand and load	4	electricity demand and system load conditions
Actual generation	6	realized supply mix by technology
Lagged generation	6	persistence in supply and generation conditions
Generation forecasts	4	day-ahead renewable generation forecasts
Autoregressive price lags	4	recent price persistence and short-run memory
Commodity prices	3	gas, oil and CO_2 cost shocks
Weather variables	2	capital-city temperature and weather conditions
Temporal indicators	8	hour, weekday, month, season, weekend and holiday effects

Table 2. The information set contains 37 variables grouped into eight predictor families; the panel contains 236,640 hourly observations.

Predictors are aligned by delivery date and realized same-day prices are excluded, so the trading test uses only information available before next-day bids are submitted.

Price distributions motivate a distributional model

Figure 5. Day-ahead price distributions by country



Market	Mean	Std	Skew	Neg. h
Romania	101.76	103.01	2.62	106
Hungary	103.71	103.46	2.60	337
Czechia	92.46	94.84	2.70	685
Bulgaria	97.97	97.55	2.52	62

Table 3. Summary statistics of hourly day-ahead prices.

- Standard deviations are close to mean prices.
- All markets are right-skewed and contain extreme positive outliers.
- Negative-price hours make the lower tail economically relevant for charging decisions.

Why a point forecast is not enough

Point forecast

A single expected price for each country and hour.

Useful for average error metrics such as MAE and RMSE.

Limitation: it hides uncertainty around rare low-price and high-price hours.

Distributional forecast

A full range of possible prices for each next-day hour.

Output: medians, quantiles, intervals and scenarios.

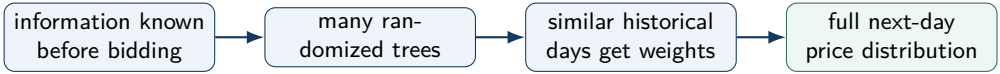
In the paper, medians select hours and quantiles set limit bid prices.

Why this matters for arbitrage

- Profit is driven by the ordering of low and high hours.
- The economically important hours are often tail events.
- A forecast can look good on average and still miss the hours that matter for storage.
- Similar arguments support probabilistic outputs in *Nowotarski and Weron (2018)*, *Cornell et al. (2024)*, *Cramer et al. (2023)* and *O'Connor et al. (2025)*.

Proper scoring-rule foundation: *Gneiting and Raftery (2007)*.

How the DRF forecast is constructed



What the forest learns

- how load, renewables, fuels, weather and calendar conditions separate different price regimes;
- where the whole price distribution changes, not only where the average changes;
- which historical days are most informative for the next delivery day.

What the model outputs

- medians used to select charge and discharge hours;
- quantiles used to set limit bid prices;
- prediction intervals for uncertainty;
- joint scenarios across 24 hours and 4 markets.

The split criterion follows *Cevid et al. (2022)* and compares distributions through FourierMMD. This avoids imposing a Gaussian shape on heavy-tailed electricity prices.

How the model is applied to our data

One forecast object

For every delivery day, the response contains all 24 hours in all 4 markets.

$$24 \times 4 = 96 \text{ prices}$$

Why joint

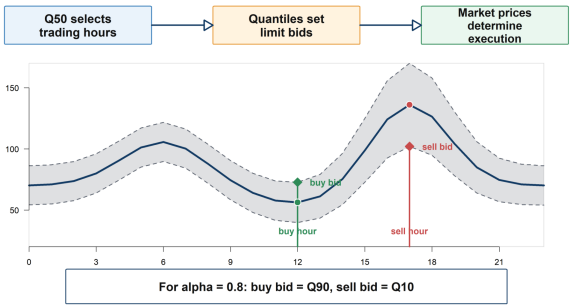
The battery compares prices within a day and, in the regional case, across markets. The joint object keeps this ordering.

Rolling out-of-sample design

- Train on the most recent 365 daily observations.
- Predict the subsequent 30-day block.
- Roll the window forward through the sample.
- Baseline forest: 500 trees and minimum terminal node size 15.
- Total evaluation: 70 rolling test windows.

From forecast distribution to day-ahead limit orders

Figure 6. Forecast distribution translated into quantile limit bids



Trading rule

- Q50 selects the charge hour h_1 and the later discharge hour h_2.
- Quantiles set buy and sell limit prices.
- Market prices determine execution.

$$q = \frac{1-\alpha}{2}$$

$$b_d^{buy} = \widehat{Q}_{1-q,d,h_1}$$

$$b_d^{sell} = \widehat{Q}_{q,d,h_2}$$

For $\alpha = 0.8$, $q = 0.1$: buy at Q90 and sell at Q10.

Adapted from the quantile-based bidding logic in *Nitka and Weron (2023)*.

Single-market and multi-market operation

Single-market case

The battery buys and sells inside one bidding zone.

This tests the stand-alone value of a battery with access to one market.

Benchmark comparison

The fixed-hour rule does not use forecasts. Perfect foresight uses realized prices and provides an upper bound.

Multi-market case

The rule selects the buying country and hour, then the selling country and later hour.

This is a stylized benchmark for regional flexibility value.

Market frictions

Country-specific tariffs are included. Cross-border fees, congestion costs and transmission-right costs are not imposed.

Evaluation metrics used in the paper

Forecast quality

MAE is the average absolute distance between the median forecast and the realized price.

RMSE gives extra weight to large mistakes, so price spikes matter more.

CRPS scores the whole predictive distribution, rewarding probability mass placed near the realized price.

Economic value

Profit is realized selling revenue minus buying cost, efficiency losses and tariffs.

NPV discounts future annual profits and subtracts the upfront investment.

IRR is the discount rate at which NPV becomes zero.

Payback is the number of years needed to recover the upfront investment.

Reference battery and financial assumptions

A 1 MW / 1 MWh unit means the battery can charge or discharge up to 1 MW and store 1 MWh of usable energy. CAPEX of EUR 500,000 is the upfront investment. Efficiency of 90 percent means that only 0.9 MWh is recovered for each 1 MWh charged. The 15-year lifetime and 7 percent discount rate convert annual trading profits into present value.

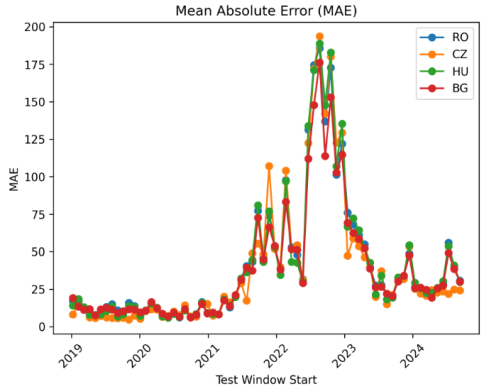
Forecasting performance is comparable across markets

Market	MAE	RMSE	CRPS
Bulgaria	37.23	46.76	26.46
Czechia	37.84	45.16	26.66
Hungary	39.81	49.26	28.15
Romania	39.63	49.60	28.12

Table 4. Average DRF forecasting performance over 70 rolling windows.

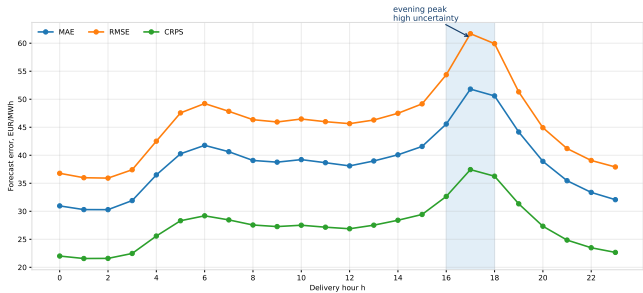
- Errors are broadly similar across countries.
- CRPS is consistently below MAE.
- Stress periods are common across the region.

Figure 7. Rolling MAE performance



Forecast errors are highest in the economically important hours

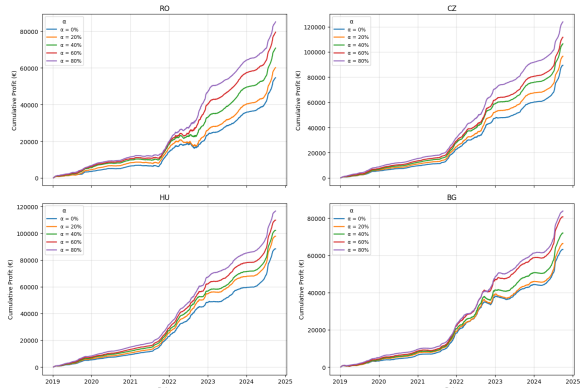
Figure 8. Hourly DRF error profile



- Overnight prices are easier to forecast.
- Errors rise during the morning ramp and peak around 17:00 to 18:00.
- Those peak hours are usually discharge hours for storage.
- The distributional forecast keeps uncertainty visible exactly where point forecasts are weakest.

Single-market trading results

Figure 9. Cumulative realized profits in the single-market setting



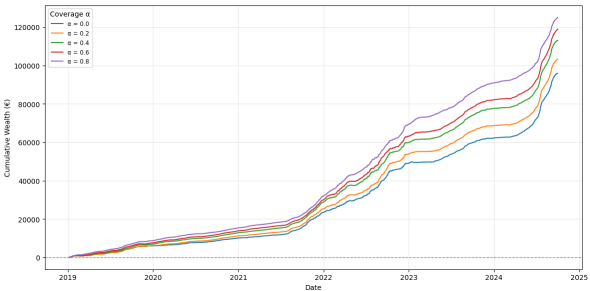
Market	Quantile	Fixed	Perfect
RO	143.0	133.2	188.2
CZ	92.1	80.8	127.7
HU	129.9	119.6	169.4
BG	137.8	122.1	179.4

Table 5. Cumulative profit in thousand EUR. Quantile strategy uses $\alpha = 0.8$.

- Profit rises with α in all four markets.
- The quantile strategy at $\alpha = 0.8$ beats the fixed-hour benchmark everywhere.
- Perfect foresight remains an upper-bound benchmark, not an implementable strategy.

Regional dispatch adds operational value

Figure 10. Cumulative realized profits in the multi-market setting



Rule	Profit	NPV	IRR	Route
Q 0.0	128.5	-295.9	-4.60	CZ→RO
Q 0.4	139.9	-277.9	-3.67	CZ→RO
Q 0.8	150.3	-261.3	-2.87	BG→RO
Fixed	134.7	-286.0	-4.08	BG→RO
Perfect	221.7	-147.9	1.91	CZ→HU

Table 6. Multi-market results. Profit and NPV in thousand EUR.

- Best forecast-driven regional profit reaches EUR 150.3k.
- The dominant route shifts from CZ→RO to BG→RO at high α.
- NPV remains negative under baseline CAPEX.

Conclusions in context

- The four CEE markets provide a useful setting because price volatility, negative-price episodes and energy-mix heterogeneity create storage opportunities.
- Distributional Random Forest is valuable because it produces a full joint next-day distribution, preserving hourly and cross-market dependence needed by the trading rule.
- The quantile-based strategy has operational value: at $\alpha = 0.8$, it beats the fixed-hour benchmark in all four single-market tests and in the multi-market benchmark.
- Regional flexibility adds value because the multi-market quantile profit exceeds the best single-market quantile result.
- The investment conclusion remains conservative: day-ahead arbitrage alone does not cover the baseline CAPEX under the paper's assumptions.